

Symmetry-Compatible Hypothesis Volume Predicts Out-of-Distribution Generalization

Subtitle. Weakness, not generic compression alone, in shortcut-compatible learning problems.

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Abstract

We study model selection in finite-domain settings where the training data admits both a local shortcut and a globally symmetry-compatible rule. In these settings, training loss and standard in-distribution validation cannot distinguish the two completions, and generic simplicity or flatness proxies can prefer the shortcut. We define a finite operationalization of Bennett-style **weakness** as the number of transformations under which a candidate function remains equivariant up to an output action. Across cyclic and dihedral symbolic task families (500 trials each), weakness selects the invariant rule in 100% of trials (Wilson 95% lower bound 0.992), while the tested train-loss, validation, description-length, MDL-style, compression, and flatness proxies select the shortcut in 100% of trials (Wilson 95% upper bound 0.008). A data-inferred selector, built by enumerating circular-domain transformations and retaining those consistent with training pairs, matches the oracle selector on cyclic and dihedral families without using the hidden offset or reflection parameter. On trained neural models, learned-function weakness under the true group is the strongest correlate of OOD accuracy across 256 local MLPs (Pearson $r = +0.817$), a 1024-model Modal replication ($r = +0.813$), and a 4096-model Modal rescale ($r = +0.8085$), while held-out validation, training loss, parameter norm, and Hutchinson sharpness remain substantially weaker. A rotated-stroke vision benchmark gives the same ordering (rotation weakness $r = +0.672$; wrong-group control $r = -0.341$). Negative cases with parity and full symmetric groups delineate the operating regime, and a Pythia-70M paraphrase experiment is reported as an exploratory latent-geometry result rather than behavioral evidence. The claim is not that compression is useless in general; it is that, in shortcut-compatible symmetry tasks, the relevant inductive bias is symmetry-compatible hypothesis volume.

1. Introduction

Modern learning pipelines are often underspecified: many predictors achieve indistinguishable training or in-distribution validation performance, yet behave differently under deployment shifts. Shortcut learning is one familiar manifestation of this problem: a local rule solves the observed benchmark but fails when the deployment distribution changes. In the experiments below, we make this ambiguity explicit. The training data is compatible with both a local shortcut and a globally transportable rule. The question is which train-perfect completion a selector should prefer.

The usual model-selection vocabulary gives several candidates: minimize training loss, prefer shorter descriptions, approximate Solomonoff or MDL priors, prefer flat minima, minimize parameter norm, or trust held-out validation. These are valuable heuristics in many settings, but they are not targeted at the particular ambiguity created by partially observed symmetries. If the deployment distribution is generated by a transformation structure, the relevant question is not only "which hypothesis is short?" but "which hypothesis remains valid under the transformations that generate the missing cases?"

This paper tests that question through **weakness**: the size of the transformation set under which a candidate function remains compatible with itself. The term comes from Bennett's compatible-world-volume framing, but the operational measure here is deliberately finite and checkable: count the group elements for which the learned function is equivariant up to an output-side action.

Contributions:

1. We define a finite-domain weakness score for candidate functions and learned neural function tables.
2. We construct shortcut-compatible symbolic tasks where the local shortcut and the invariant rule are both train-perfect, but only the invariant generalizes to OOD inputs.
3. We show that weakness selects the OOD-correct completion on cyclic and dihedral families where the candidate group has the right granularity, while the tested loss, validation, description-length, compression, MDL-style, and flatness proxies do not.
4. We verify the neural version across 256 local MLPs, a 1024-model Modal replication, and a 4096-model Modal rescale.
5. We report honest boundaries: parity is too small a group to disambiguate, the full symmetric group is too uninformative for this candidate pool, and the language result currently supports only a latent-geometry claim.

Claim level. This is evidence for a symmetry-volume model-selection principle in finite shortcut-compatible regimes. It is not a claim that generic compression, PAC-Bayes, validation, or flatness never matter. A safer summary is: when train-perfect shortcuts and invariant rules coexist, generic simplicity proxies can be underdetermined or misaligned, while symmetry-compatible volume directly measures the completion that will transport across the group-generated OOD split.

2. Definitions and Selection Principle

Let $f: X \rightarrow X$ be a candidate function on a finite domain X of size n . Let G be a finite group or finite transformation set acting on X .

With-action equivariance. We say f is compatible with $g \in G$ if there exists an output-side action $h \in G$ such that

$$f(g \cdot x) = h \cdot f(x) \quad \forall x \in X.$$

This generalizes strict equivariance ($h = g$) and is useful when input and output actions are related but not necessarily identical.

Weakness.

$$W_G(f) = \left| \left\{ g \in G : \exists h \in G, \forall x, f(g \cdot x) = h \cdot f(x) \right\} \right|.$$

A function with $W_G(f) = |G|$ is fully compatible with the candidate transformation set. A function with $W_G(f) = 1$ is compatible only with the identity.

Weakness selector. Given a finite candidate pool $f_1, \dots, f_K$ and a candidate transformation set $\hat{G}$, the weakness selector returns

$$f_{i^*}, i^* \in \operatorname{argmax}_i W_{\hat{G}}(f_i),$$

with ties broken by the same compression proxy used by the baseline selectors.

Operational selection principle. If the training data constrains only part of a transformation orbit, then a train-consistent hypothesis that remains compatible with more elements of the deployment-generating transformation set should cover more of the missing orbit. The experiments below test this principle in regimes where the shortcut and invariant are both train-perfect, so the comparison is not confounded by ordinary training error.

The principle has a clear failure mode. If $\hat{G}$ is too small, many wrong local completions can be equally

compatible. If $\hat{G}$ is too large or semantically uninformative, wrong candidates may have comparable compatibility counts. Those are not footnotes; they are part of the proposed operating regime.

3. Symbolic Benchmark

3.1 Task Families

We use four families. Each trial contains a finite domain, a ground-truth function, a biased training subset, an OOD subset, and a candidate pool containing at least a local shortcut and the true invariant.

- **cyclic_prefix_shift**: domain Z_n , $n \in 7, 11, 13$. Truth $f(x) = (x + b) \bmod n$. Training observes a prefix window $w \in 3, 4, 5$; OOD inputs are the suffix.
- **dihedral_reflection**: domain Z_n , $n \in 7, 9, 11, 13$. Truth $f(x) = (b - x) \bmod n$. Training observes a prefix window; OOD inputs are the remaining indices. The candidate transformation family is D_n .
- **parity_coset**: even domain size $n \in 6, 8, 10$. Truth swaps each parity partner $x \mapsto x \oplus 1$. Training observes one parity coset; the other coset is OOD.
- **color_permutation**: domain size $n \in 4, 5, 6$. Truth is a sampled non-identity permutation. Training observes a sparse subset. The candidate transformation family is S_n .

The candidate pool always includes a local patch that copies the observed training outputs and defaults to identity elsewhere, a memorizer, the true invariant, and available wrong invariants that also fit the training pairs. The local patch is intentionally short; this makes the compression comparison adversarial rather than decorative.

3.2 Selectors and Baselines

We compare eleven selectors. In the discussion below, "classical baselines" means these implemented proxies, not every possible MDL, PAC-Bayes, flatness, or validation method.

Selector	Definition
train_loss	best training accuracy, ties broken by simplicity
validation	leave-one-out accuracy on observed training pairs
simplicity	shortest hand-coded hypothesis form length
compression	<code>form_length + 20 * train_errors</code>
mdl_program	$2^{\text{form_length}}$, a Solomonoff-style proxy
flatness_proxy	count of domain positions unconstrained by training
weakness_oracle	maximum weakness under the true candidate group
weakness_wrong_group	weakness under random non-cyclic permutations
weakness_noisy_group	weakness under a 50%-corrupted subset of the true group
weakness_data_inferred	weakness under transformations retained by training-pair consistency
random	uniform random train-consistent candidate

The validation baseline is deliberately in-distribution: it holds out one observed training pair and scores the

fixed candidate on that pair. It is not allowed to see OOD transformations. Wilson intervals use the standard score interval with $z = 1.959963984540054$.

The `weakness_data_inferred` selector does not receive the hidden offset, reflection parameter, or ground-truth function. For cyclic tasks it enumerates circular shifts and keeps shifts that do not contradict observed pairs under some output shift. For dihedral tasks it enumerates the signed circular prior (rotations plus reflections) and keeps transformations consistent with the observed pairs under some output-side signed action. For color permutation, the current heuristic falls back to a small cyclic prior and is intentionally weak.

3.3 Results

Each family uses 500 independent trials. Wilson 95% intervals are shown for invariant-recovery rates.

`cyclic_prefix_shift`

Selector	Invariant rate	Wilson 95% CI	Mean OOD
<code>weakness_oracle</code>	1.000	(0.992, 1.000)	1.000
<code>weakness_data_inferred</code>	1.000	(0.992, 1.000)	1.000
<code>weakness_noisy_group</code>	1.000	(0.992, 1.000)	1.000
<code>random</code>	0.346	(0.306, 0.389)	0.380
<code>weakness_wrong_group</code>	0.002	(0.000, 0.011)	0.003
<code>train_loss, simplicity, compression, mdl_program, flatness_proxy, validation</code>	0.000	(0.000, 0.008)	0.000

`dihedral_reflection`

Selector	Invariant rate	Wilson 95% CI	Mean OOD
<code>weakness_oracle</code>	1.000	(0.992, 1.000)	1.000
<code>weakness_data_inferred</code>	1.000	(0.992, 1.000)	1.000
<code>weakness_noisy_group</code>	0.972	(0.954, 0.983)	0.976
<code>random</code>	0.340	(0.300, 0.383)	0.413
<code>weakness_wrong_group</code>	0.018	(0.009, 0.034)	0.119
<code>classical baselines</code>	0.000	(0.000, 0.008)	0.107

`color_permutation` (partial positive)

Selector	Invariant rate	Wilson 95% CI	Mean OOD
weakness_noisy_group	0.858	(0.825, 0.886)	0.890
weakness_oracle	0.824	(0.788, 0.855)	0.875
random	0.322	(0.283, 0.364)	0.455
weakness_data_inferred	0.138	(0.111, 0.171)	0.325
classical baselines, weakness_wrong_group	0.000	(0.000, 0.008)	0.210

parity_coset (negative case)

Selector	Invariant rate	Mean OOD
random	0.298	0.341
weakness_wrong_group	0.038	0.038
weakness_noisy_group	0.022	0.022
weakness_oracle, weakness_data_inferred, classical baselines	0.000	0.000

Cyclic and dihedral exhibit the intended separation: the implemented generic baselines choose the local shortcut, while weakness chooses the invariant. The color-permutation result is a partial positive: even the oracle S_n score is informative, but the data-inferred cyclic prior is not enough to recover the hidden permutation structure. Parity is a clean negative: $|G| = 2$ is too small to separate the truth from train-perfect local patches that are also parity-compatible.

4. Neural Weakness as a Predictor of OOD Accuracy

4.1 Setup

We train small MLPs on cyclic-prefix-shift tasks with modulus 11 and training window 3. Hyperparameters vary across:

- depth $\in \{1, 2, 3\}$, hidden width $\in \{16, 32, 64, 128\}$, initialization scale $\in \{0.3, 0.7, 1.0, 1.5\}$;
- optimizer $\in \{\text{Adam}, \text{SGD}+\text{momentum}\}$, learning rate $\in \{10^{-3}, 3 \cdot 10^{-3}, 10^{-2}, 3 \cdot 10^{-2}\}$, weight decay $\in \{0, 10^{-4}, 10^{-2}\}$;
- augmentation $\in \{\text{none}, \text{partial cyclic}, \text{full cyclic}, \text{wrong reflection}, \text{wrong random}\}$.

After 2000 steps, we extract the full argmax function table $\hat{f}$ and compute training loss, held-out validation accuracy on observed-support examples, parameter L_2 , Hutchinson sharpness proxy, true-group weakness, partial-cyclic weakness, wrong-random-permutation weakness, and random-label weakness. The dependent variable is OOD accuracy on the held-out suffix.

4.2 Local 256-MLP Sweep

Across 256 trained MLPs, mean OOD accuracy is 0.3335 and 23.8% of models reach perfect OOD accuracy.

Predictor	Pearson r	Spearman rho
weakness_oracle_norm	+0.817	+0.552
weakness_partial_cyclic_norm	+0.804	+0.540
weakness_oracle (raw)	+0.763	+0.715
Hutchinson sharpness proxy	+0.129	+0.142
parameter L_2	+0.099	+0.308
held-out validation accuracy	+0.096	+0.058
training loss	-0.031	+0.136
weakness_wrong_group_norm	-0.129	-0.057
weakness_random_label_norm	-0.116	-0.051

Per-augmentation breakdown:

Augmentation	n	Mean OOD	Mean weakness
full_cyclic	54	0.939	0.951
partial_cyclic	48	0.618	0.320
wrong_random	50	0.084	0.140
wrong_reflection	50	0.017	0.117
none	54	0.000	0.141

4.3 Modal Replications

The 1024-model Modal replication uses 8 shards by 128 models with the same hyperparameter space and base seed family. The 4096-model rescale uses 32 shards by 128 models with base seed 20260702. Both preserve the ordering.

Predictor	256 Pearson	1024 Pearson	4096 Pearson	4096 Spearman
weakness_oracle_norm	+0.817	+0.813	+0.8085	+0.5417
weakness_partial_cyclic_norm	+0.804	+0.804	+0.7940	+0.5343
parameter L_2	+0.099	+0.273	+0.2533	+0.3313
held-out validation accuracy	+0.096	+0.089	+0.0924	+0.0427
Hutchinson sharpness proxy	+0.129	+0.134	+0.0848	+0.1617
training loss	-0.031	-0.048	-0.0249	+0.1200

weakness_wrong_group_norm	-0.129	-0.116	-0.1040	-0.0218
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4096-model per-augmentation breakdown:

Augmentation	n	Mean OOD	Mean weakness
full_cyclic	805	0.9434	0.9545
partial_cyclic	806	0.6438	0.3678
wrong_random	841	0.0877	0.1230
wrong_reflection	823	0.0135	0.1315
none	821	0.0000	0.1157

The wrong-group and random-label controls rule out the trivial explanation that any equivariance count predicts OOD. The observed signal is specific to the group aligned with the deployment shift.

5. Wrong-, Noisy-, and Data-Inferred-Group Ablations

The benchmark exposes a precise operating regime:

- **Wrong group:** random non-cyclic permutations select the invariant in only 0.2% of cyclic trials and 1.8% of dihedral trials.
- **Noisy group:** a transformation set containing roughly half the true group plus one wrong element still recovers the invariant in 100% of cyclic and 97.2% of dihedral trials.
- **Data-inferred group:** consistency filtering over the circular-domain prior recovers the invariant in 100% of cyclic and dihedral trials. This uses the domain's enumerable transformation prior and observed pair consistency, not the hidden offset, hidden reflection parameter, or ground-truth function.
- **Too small:** parity fails because the group is not rich enough to separate the truth from local patches.
- **Too large/unstructured:** S_n is informative for the oracle selector but difficult for the current data-inferred heuristic; wrong involutions can have comparable centralizer-orbit sizes.

These failures matter. Weakness is not a universal scalar that solves OOD generalization. It becomes load-bearing when the candidate transformation set is aligned with the deployment-generating structure and has enough resolution to distinguish train-perfect completions.

6. Scaling Beyond Cyclic Symbolic Tasks

6.1 Vision: Rotated-Stroke Partial-Orbit Supervision

We construct a synthetic stroke-classification task with eight classes on 16 by 16 grayscale images. The underlying transformation is Z_8 rotation. Training shows each class at only three of the eight rotation angles; the remaining angles are OOD. This mirrors Perin and Deny's partially observed cyclic-symmetry setup, with neural weakness added as a measurement.

Across 96 CNN/MLP models:

Predictor	Pearson r	Spearman rho
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weakness_rotation_norm	+0.672	+0.573
weakness_wrong_group_norm	-0.341	-0.452
parameter L_2	+0.333	+0.269
train accuracy	+0.332	+0.265
final training loss	-0.319	-0.221
Hutchinson sharpness	+0.198	+0.368

Per-augmentation breakdown:

Augmentation	n	Mean OOD	Mean weakness
partial_rotation	25	0.734	0.776
wrong_permute	20	0.264	0.388
none	28	0.275	0.402
full_rotation	23	0.272	0.432

The wrong-group control is anti-correlated, consistent with a real tradeoff between rotation-compatible structure and pixel-permutation-compatible structure.

6.2 Language: Pythia-70M Paraphrase Invariance

We extract per-layer mean-pooled hidden states from Pythia-70M for 24 concepts with 3 paraphrase variants each, treating each concept's variants as an approximate paraphrase orbit. We compute same-orbit cosine, wrong-orbit cosine, and next-token argmax consistency. Raw cosine is dominated by embedding anisotropy, so we also apply per-layer mean-centering following All-but-the-Top.

After centering, same-orbit paraphrase similarity exceeds the wrong-orbit control by +0.44 to +0.79 across layers:

Layer	Same-orbit centered	Wrong-orbit centered	Gap
0	0.400	-0.038	+0.438
1	0.568	-0.020	+0.588
2	0.565	-0.030	+0.596
3	0.718	+0.029	+0.689
4	0.742	+0.008	+0.734
5	0.726	-0.067	+0.793
6	0.700	-0.014	+0.713

This is a real latent-geometry signal, but it does not predict per-concept next-token behavioral consistency

at this scale: Pearson(centered weakness, behavior) lies in $[-0.35, -0.13]$ across layers with only 24 concepts. We therefore treat the language section as exploratory. It supports "paraphrase orbits cluster in centered latent space," not "latent weakness predicts LLM behavior."

7. Related Work

- **Shortcut learning and underspecification.** Geirhos et al. frame shortcuts as rules that solve benchmarks but fail under changed test conditions. D'Amour et al. show that standard held-out validation can leave pipelines underspecified, returning predictors with similar validation performance but different deployment behavior. Our benchmark is a controlled finite-domain version of this ambiguity.
- **Geometric deep learning and equivariance.** Cohen and Welling, Kondor and Trivedi, and Bronstein et al. formalize how architectural equivariance builds transformation structure into the model class. Our contribution is a measurement/selection criterion for candidate functions and learned function tables, rather than a new equivariant architecture.
- **Symmetry learning from data.** Perin and Deny analyze partially observed cyclic orbits and show when conventional supervised networks fail to generalize symmetries not present in the architecture or sufficiently dense in the data. Van der Ouderaa, Immer, and van der Wilk learn layer-wise equivariances using gradients and model evidence. Our data-inferred selector is much simpler and enumerative, but it targets the same gap: useful symmetries are often not known in advance.
- **Simplicity, MDL, and function-space priors.** Solomonoff and Hutter provide the algorithmic-probability foundation; Valle-Perez, Camargo, and Louis argue that neural parameter-function maps are biased toward simple functions. Our result is not a refutation of simplicity bias. It shows that, in these shortcut-compatible constructions, the tested generic simplicity proxies prefer the wrong train-perfect completion.
- **Flatness and sharpness.** Hochreiter and Schmidhuber and Keskar et al. motivate flat minima; Dinh et al. show that sharpness can be changed by reparameterization without changing the represented function. Bennett's recent flat-minima critique argues for weakness as a function-level alternative. Our Hutchinson proxy is only one flatness measure, so the empirical claim is deliberately narrow.
- **Grokking and emergence of structure.** Power et al. and Liu et al. document transitions from memorization to structured generalization. The neural sweeps here give one concrete variable, learned-function weakness, that tracks the structured side of that transition in cyclic tasks.

8. Limitations and Negative Results

1. The symbolic benchmark uses small finite domains. Exact weakness counting will not scale to large or continuous groups without generator-based approximations or sampling.
2. The data-inferred selector is not open-ended group discovery. It uses an enumerable transformation prior for the domain and filters it by training-pair consistency.
3. The MDL, compression, validation, norm, and flatness baselines are intentionally simple proxies. Stronger PAC-Bayes, compression, or validation schemes could be tested and should be.
4. The neural tasks are synthetic and small. The 4096-model rescale improves statistical stability but does not make the cyclic task naturalistic.
5. The vision task is still synthetic, although it is closer to standard partial-orbit image settings than the symbolic benchmark.
6. The language result is latent-geometric only. It should not be interpreted as behavioral paraphrase robustness in LLMs.
7. Weakness fails when the transformation set is too small, too large, or not aligned with the deployment shift. Those failures are part of the theory's boundary, not exceptions to hide.

9. Reproducibility

All core code is in `experiments/symbolic_weakness/`. The committed result reports are:

- `experiments/symbolic_weakness/results/multi_family_500_2026_06_09.md`
- `experiments/symbolic_weakness/results/neural_sweep_v3_2026_06_09.md`
- `experiments/symbolic_weakness/results/modal_neural_sweep_v1_2026_06_09.md`
- `experiments/symbolic_weakness/results/modal_neural_sweep_2026_07_02.md`
- `experiments/rotation_weakness/results/sweep_v1_2026_06_09.md`
- `experiments/paraphrase_weakness/results/pythia_70m_2026_06_09.md`

Commands:

```
# Symbolic multi-family benchmark
python3 -m experiments.symbolic_weakness.benchmark \
  --trials-per-family 500 --seed 20260609 \
  --out artifacts/symbolic_weakness/multi_family_500.json

# Local neural sweep
python3 -m experiments.symbolic_weakness.neural \
  --n-models 256 --epochs 2000 --base-seed 20260609 \
  --out artifacts/symbolic_weakness/neural_sweep_v3.json

# 1024-model Modal neural sweep
doppler --scope /Users/jawaun/superoptimizers run -- \
  uvx --python 3.12 --from modal modal run \
  experiments/symbolic_weakness/modal_neural_sweep.py \
  --n-shards 8 --models-per-shard 128 --epochs 2000 \
  --base-seed 20260609 \
  --out artifacts/symbolic_weakness/modal_neural_sweep_v1.json

# 4096-model Modal rescale
doppler --scope /Users/jawaun/superoptimizers run -- \
  uvx --python 3.12 --from modal modal run \
  experiments/symbolic_weakness/modal_neural_sweep.py \
  --n-shards 32 --models-per-shard 128 --epochs 2000 \
  --base-seed 20260702 \
  --out artifacts/symbolic_weakness/modal_neural_sweep_2026_07_02.json
```

Quality gates:

```
python3 -m unittest discover -s tests -p "test_symbolic*"
python3 -m unittest discover -s tests -p "test_rotation*"
python3 scripts/regen.py symbolic_weakness
```

10. Discussion

The discriminating quantity in these experiments is not generic shortness alone. It is compatibility with the transformations that generate the missing cases. When a local shortcut and a global invariant are both train-perfect, loss and in-distribution validation are silent; the tested simplicity proxies prefer the shorter local patch; and parameter-space diagnostics are at best indirect. Weakness directly scores the structure that matters for the OOD split.

The data-inferred result is the practical version of the symbolic claim. The selector does not need the hidden

offset or hidden reflection; it needs an enumerable transformation prior and enough observed pair structure to avoid contradiction. This is still far from open-ended symmetry discovery, but it is a meaningful step away from oracle-group scoring.

The negative cases should guide future work. Parity says that some transformation sets are too coarse to select the transportable rule. The S_n case says that a huge, weakly structured group can make many wrong completions look similarly compatible. The language result says that latent clustering is not behavior. These boundaries make the claim more useful: weakness is a model-selection variable for regimes where the candidate transformation set is aligned, granular, and behaviorally relevant.

The next steps are clear: replace enumeration with learned transformation generators, test stronger compression and PAC-Bayes baselines, scale from synthetic groups to natural partial-orbit datasets, and turn weakness from a post-hoc selector into a training-time regularizer.

Acknowledgements and Code

The benchmark, neural sweep, Modal endpoint, summarization tools, and tests are available at <https://github.com/jawaunb/research-derived-experiments> under `experiments/symbolic_weakness/`.

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